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[MSCCH104]

FIRST SEMESTER M.Sc. (CHEMISTRY) EXAMINATION, DECEMBER 2015

COMPUTATIONAL CHEMISTRY, MATHEMATICS FOR CHEMISTS, ENVIRONMENTAL CHEMISTRY AND GREEN CHEMISTRY

Time : 03 Hours **Maximum Marks:80**

Instructions:
1. Questions **1 to 6** carry **15** marks each. Answer any FOUR.
2. Question No. **7** is compulsory carries **20** marks.

1. **a.** Explain the characteristics of a computer.

b. Explain different types of logic gates.

c. Answer the following;

i. Convert $79_{(10)}$ to octal. **ii.** Convert $AD_{(16)}$ to decimal. **4+5+6**

2. **a.** Find the inverse of a matrix $\begin{bmatrix} 3 & -2 \\ -6 & 4 \end{bmatrix}$

b. Find the derivative of $(sinx)^{logx}$.

c. Find the maximum and minimum value of the function $2x^3 - 3x^2 - 36x + 10$.

d. A urn contains 2 white and 2 black balls and a second urn contains 2 white and 4 black balls. If a ball is drawn from each find the probability that:

i. Both are white **ii.** Both are black. **4+4+4+3**

3. **a.** Give an account of global water availability and usage.

b. Explain in detail about water quality parameters and standards.

c. What are water induced diseases? Explain with examples. **5+5+5**

4. **a.** State the law of conservation of energy and explain by considering oscillating simple pendulum as an example.

b. Write equations for kinetic and potential energies, how fast should a man of mass 50 kg run, so that his kinetic energy is 625 J?

c. Explain the working principle of hydrogen fuel cells. **5+5+5**

5. **a.** Explain the following green chemistry principles with suitable examples,

i. Pollution prevention **ii.** Atom economy

b. Discuss the solvent free Michael addition reaction with suitable examples.

c. Write a note on green organic oxidation reactions. **5+5+5**

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5. a. Discuss the advantages of microwave assisted organic synthesis of benzoic acid over conventional method.

b. Write a note on ultrasound assisted green synthesis.

c. Explain the Stetter reaction with mechanism. **5+5+5**

7. Answer any ***FOUR*** of the following.

a. Fit a second degree polynomial of the form $y = a + bx + cx2$ for the data:

x	0	1	2	3	4	5
y	1	3	7	13	21	31

b. Explain the universal gates.

c. Write a note on carbon cycle.

d. Compare nuclear fission and nuclear fusion.

e. Discuss with suitable examples, the rules for selection of an appropriate solvent which favours the green organic synthesis.

f. Explain Robinson annulation reaction with suitable examples. **5+5+5+5**